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$$1) a) \sum_{n=1}^{\infty} \frac{2}{3^n} = \frac{2}{3} + \frac{2}{9} + \dots + \frac{2}{3^n} + \dots$$

$$q = \frac{1}{3} ; \quad \frac{2^n - 1}{2^{-1}} = \sum_{k=0}^{n-1} 2^k = 1 + 2 + 2^2 + \dots + 2^{n-1}$$

$$n \rightarrow \infty \Rightarrow \frac{-1}{2^{-1}} = \frac{1}{1-2} = 1 + 2 + 2^2 + \dots$$

$$\frac{2}{3} + \frac{2}{9} + \dots = \frac{2}{3} \cdot \underbrace{\left(1 + \frac{1}{3} + \dots\right)}_{\frac{1}{1-\frac{1}{3}}} = \frac{2}{3} \cdot \frac{1}{1-\frac{1}{3}} =$$

$$= \frac{2}{3} \cdot \frac{1}{\frac{2}{3}} = 1$$

$$b) \sum_{n=1}^{\infty} \frac{2n+1}{n!} = \frac{2+1}{1!} + \frac{4+1}{2!} + \dots + \frac{2n+1}{n!} + \dots \Rightarrow$$

$$\sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots = e$$

!!

$$\Rightarrow \sum_{n=1}^{\infty} \frac{2n}{n!} + \sum_{n=1}^{\infty} \frac{1}{n!} = 2 \cdot \sum_{n=1}^{\infty} \frac{1}{(n-1)!} + e - 1 =$$

$$= 2e + e - 1 = 3e - 1$$

$$c) \sum_{n \geq 1} \frac{1}{4n^2 - 1} = \sum_{n \geq 1} \frac{1}{(2n-1)(2n+1)} =$$

$$= \frac{1}{2} \sum_{n \geq 1} \frac{(2n+1) - (2n-1)}{(2n-1)(2n+1)} =$$

$$= \frac{1}{2} \sum_{n \geq 1} \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right)$$

$$= \frac{1}{2} \left(1 - \cancel{\frac{1}{3}} + \cancel{\frac{1}{3}} - \cancel{\frac{1}{5}} + \dots \right)$$

$$= \frac{1}{2} \lim_{n \rightarrow \infty} \left(1 - \cancel{\frac{1}{3}} + \cancel{\frac{1}{3}} - \cancel{\frac{1}{5}} + \dots + \cancel{\frac{1}{2n-1}} - \underbrace{\frac{1}{2n+1}}_{\rightarrow 0} \right)$$

$$= \frac{1}{2}$$

$$d) \sum_{n \geq 1} \frac{1}{n(n+1)(n+2)}$$

$$\frac{1}{n(n+1)(n+2)} = \frac{\overset{n+2}{\cancel{1}}}{n(n+1)} - \frac{\overset{n+1}{\cancel{1}}}{n(n+2)}$$

$$= \underbrace{\sum_{n \geq 1} \frac{1}{n(n+1)}}_{=1} - \underbrace{\sum_{n \geq 1} \frac{1}{n(n+2)}}_{=1} \quad (1)$$

$$\sum_{n \geq 1} \frac{1}{n(n+1)} = \sum_{n \geq 1} \frac{n+1-n}{n(n+1)} =$$

$$= \sum_{n \geq 1} \frac{\cancel{n+1}}{n(n+1)} - \frac{n}{n(n+1)} = \sum_{n \geq 1} \frac{1}{n} - \frac{1}{n+1} =$$

$$= \lim_{n \rightarrow \infty} \left(1 - \cancel{\frac{1}{2}} + \cancel{\frac{1}{2}} - \frac{1}{3} + \dots + \cancel{\frac{1}{n}} - \underbrace{\frac{1}{n+1}}_0 \right)$$

$$= 1 - 0 = 1$$

$$\sum_{n \geq 1} \frac{1}{n(n+2)} = \frac{1}{2} \sum_{n \geq 1} \frac{n+2-n}{n(n+2)} =$$

$$= \frac{1}{2} \sum_{n \geq 1} \left(\frac{1}{n} - \frac{1}{n+2} \right) = \frac{1}{2} \lim_{n \rightarrow \infty} \left(1 - \cancel{\frac{1}{3}} + \cancel{\frac{1}{2}} - \cancel{\frac{1}{4}} + \cancel{\frac{1}{5}} - \right.$$

$$\left. - \frac{1}{5} + \cancel{\frac{1}{6}} - \frac{1}{6} + \dots + \cancel{\frac{1}{n}} - \frac{1}{n+2} \right) = \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}$$

$$(1) \quad 1 - \frac{3}{4} = \frac{1}{4}$$

$$e) \sum_{n \geq 1} \frac{n}{2^n} = \left(\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \dots + \frac{n}{2^n} + \dots \right)$$

$$\sum x_n, \quad \frac{x_{n+1}}{x_n} = l \quad \begin{cases} \text{if } l < 1, \sum x_n \text{ conv.} \\ \text{if } l > 1, \sum x_n \text{ div.} \end{cases}$$

$$\frac{x_{n+1}}{x_n} = \frac{n+1}{2^{n+1}} \cdot \frac{2^n}{n} = \frac{n+1}{2n} \rightarrow \frac{1}{2} < 1$$

the series is convergent

$$S = \sum_{n \geq 1} x_n = \sum_{n \geq 1} \frac{n}{2^n} = \sum_{n \geq 1} \frac{n+1-1}{2^n} =$$

$$= \sum_{n \geq 1} \frac{n+1}{2^n} - \sum_{n \geq 1} \frac{1}{2^n}$$

$$= 2 \cdot \sum_{n \geq 1} \frac{n+1}{2^{n+1}} - \underbrace{\sum_{n \geq 1} \frac{1}{2^n}}_{\frac{1}{2} + \frac{1}{4} + \dots = 1}$$

$$= 2 \sum_{n \geq 1} \frac{n+1}{2^{n+1}} - 1 = 2 \cdot \sum_{n \geq 2} \frac{n}{2^n} - 1 =$$

$$\begin{aligned}
 &= 2 \cdot \left(S - \frac{1}{2} \right) - 1 \\
 &= 2S - 2 \quad \text{excluded term}
 \end{aligned}$$

$$S = 2S - 2 \Rightarrow \boxed{S = 2}$$

General method

$$S_m = \sum_{k=1}^m k \cdot x^k = x \sum_{k=1}^m k x^{k-1}$$

$$= x \sum_{k=1}^m (x^k)' = x \cdot \left(\sum_{k=1}^m x^k \right)'$$

$$\begin{aligned}
 \sum_{k=1}^m x^k &= x + x^2 + \dots + x^m = x(1 + x + x^2 + \dots + x^{m-1}) \\
 &= x \frac{x^m - 1}{x - 1} = \frac{x^{m+1} - x}{x - 1}
 \end{aligned}$$

$$\left(\sum_{k=1}^m x^k \right)' = \left(\frac{x^{m+1} - x}{x - 1} \right)'$$

$$= \frac{(x^{m+1} - x)'(x - 1) - (x^{m+1} - x)(x - 1)'}{(x - 1)^2}$$

$$= \frac{[(m+1) \cdot x^m] (x - 1) - (x^{m+1} - x) \cdot 1}{(x - 1)^2}$$

$$= \frac{n \cdot x^{n+1} - (n+1) \cdot x^n + 1}{(x-1)^2} \xrightarrow{n \rightarrow \infty} \frac{1}{(x-1)^2}$$

$x^n \rightarrow 0$ if $|x| < 1$

$$\lim_{n \rightarrow \infty} S_n = \frac{x}{(x-1)^2} = \sum_{n=1}^{\infty} n x^n$$

ⓔ $x = \frac{1}{2} \Rightarrow \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 2$

$$f) \sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{3}}}$$

$$\sqrt[2]{n} < n \Rightarrow \frac{1}{\sqrt[3]{n}} > \frac{1}{n}$$

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty \Rightarrow \sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n}} = \infty$$

3) a) $\sum_{n=2}^{\infty} \frac{1}{\ln n}$

$$n > \ln n \Rightarrow \frac{1}{\ln n} > \frac{1}{n} \Rightarrow \sum_{n=2}^{\infty} \frac{1}{\ln n} = \infty$$

$$\sum_{n \geq 2} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$b) \sum_{n=1}^{\infty} \frac{1}{n\sqrt{n+1}}$$

$$n\sqrt{n+1} < n^2$$

$$\frac{1}{n\sqrt{n+1}} > \frac{1}{n^2} ; \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ converges}$$

$$n\sqrt{n+1} > n\sqrt{n}$$

$$\frac{1}{n\sqrt{n+1}} < \frac{1}{n\sqrt{n}}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ conv. when } p > 1$$

$$\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}} \xrightarrow{\frac{3}{2} > 1} \text{ converges} \Rightarrow$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n\sqrt{n+1}} \text{ converges}$$

comparaison

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{n\sqrt{n+1}}}{\frac{1}{n^{\frac{3}{2}}}} = 1$$

$\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = l \in (0, +\infty)$, then $(x_n), (y_n)$
have the same nature

$$c) \sum_{n=1}^{\infty} \frac{\ln(1 + \frac{1}{n})}{n}$$

$$\left(1 + \frac{1}{n}\right)^n \rightarrow e$$

$$n \ln\left(1 + \frac{1}{n}\right) \rightarrow 1$$

$$\lim_{n \rightarrow \infty} \frac{\ln\left(1 + \frac{1}{n}\right)}{\frac{1}{n}} \rightarrow 1, \quad \ln\left(1 + \frac{1}{n}\right) \approx \frac{1}{n}$$

$$\sum_{n=1}^{\infty} \frac{\ln\left(1 + \frac{1}{n}\right)}{n}, \quad \frac{\ln\left(1 + \frac{1}{n}\right)}{n} \approx \frac{1}{n^2}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \text{ converges } (2 > 1) \Rightarrow \sum_{n=1}^{\infty} \frac{\ln\left(1 + \frac{1}{n}\right)}{n} \text{ conv.}$$

$$d) \sum \frac{n!}{n^n}$$

$$\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = \lim_{n \rightarrow \infty} \frac{\frac{(n+1)!}{(n+1)^{n+1}}}{\frac{n!}{n^n}} = \lim_{n \rightarrow \infty} \frac{(n+1)}{(n+1)^{n+1}} \cdot \frac{n^n}{1} =$$

$$= \lim_{n \rightarrow \infty} \frac{(n+1) \cdot n^n}{(n+1)^{n+1}} = \frac{n^n}{(n+1)^n} = \left(\frac{n}{n+1} \right)^n = \frac{1}{\left(1 + \frac{1}{n} \right)^n}$$

$$= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right)^n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right)^n =$$

$$= e^{\lim_{n \rightarrow \infty} -\frac{n}{n+1}} = e^{-1} = \frac{1}{e} < 1 \Rightarrow$$

$$\Rightarrow \sum \frac{n!}{n^n} \text{ conv.}$$

$$e) \sum \left(\frac{n}{n+1} \right)^{n^2}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{x_n} = l < \begin{cases} l > 1 \Rightarrow \text{conv.} \\ l < 1 \Rightarrow \text{div.} \end{cases}$$

$$\sqrt[n]{X_n} = \left(\frac{n}{n+1}\right)^n \rightarrow \frac{1}{e} < 1 \Rightarrow \text{conv.}$$

$$f) \sum_{n \geq 2} \frac{1}{n \ln n} \quad \text{"like"} \quad \sum_{n \geq 1} 2^n \cdot \frac{1}{2^n \cdot \ln 2^n}$$

$$\text{Ratio test: } \frac{n \ln n}{(n+1) \ln(n+1)} \rightarrow 1 \quad ? \text{ (not helpful)}$$

Cauchy Condensation Test:

$$\sum_{n \geq 1} X_n \quad \text{"like"} \quad \sum_{n \geq 0} 2^n X_{2^n}$$

$$\hookrightarrow \sum_{n \geq 0} \frac{1}{n \ln 2} = \frac{1}{\ln 2} \sum_{n \geq 0} \frac{1}{n} = \infty \Rightarrow$$

$\Rightarrow$ our series is also divergent.